

Finetuning ASR Models on Children’s Speech

Advait Paranjpe¹, Farrah Galal¹

¹Department of Electrical and Computer Engineering, UCLA

advaitparanjpe@ucla.edu, fgala1@g.ucla.edu

Abstract

We evaluated Whisper and Wav2Vec2 on children’s speech from the OGI Kids dataset. The study compared zero-shot performance, fine-tuning, hyperparameter sweeps, training-set size, age-group transfer, and augmentation. Whisper-base was the strongest overall model in the main comparison, while Wav2Vec2 improved substantially after stabilized fine-tuning. The strongest domain effect was age: both model families performed much better on older children and struggled most on ages 4–6. After correcting a fixed-`max_steps` confound with matched-exposure reruns, the Wav2Vec2 results showed that more data helps and that training on all ages gives the best cross-age generalization. Augmentation provided limited benefit in our setup: a simple noise-plus-gain pipeline hurt performance, while VTLP gave only a slight improvement for Wav2Vec2.

Index Terms: children’s speech, ASR, Whisper, Wav2Vec2, domain adaptation

1. Introduction

This class project studies domain adaptation for children’s automatic speech recognition (ASR). Although large pretrained speech foundation models perform well on general speech, children’s speech remains challenging because of differences in pitch, pronunciation, speaking rate, and consistency relative to adult speech. We therefore evaluated how well two modern ASR model families, Whisper and Wav2Vec2, adapt to the OGI Kids corpus, and which experimental factors most strongly affect word error rate (WER).

2. Background

The OGI Kids corpus contains prompted and spontaneous speech from approximately 1100 children spanning kindergarten through grade 10 [1, 2]. This makes it a useful benchmark for studying age-dependent ASR performance. Whisper is a supervised encoder-decoder Transformer trained on large-scale web audio [3], while Wav2Vec2 is a self-supervised speech model that is later fine-tuned for ASR [4]. Because these model families are pretrained differently, it is reasonable to expect them to respond differently when adapted to children’s speech.

Augmentation is also relevant. Prior speech work shows that augmentation can improve robustness when it matches the target variability [5]. However, mismatched augmentation can fail to help or can even hurt performance, which is especially important in a child-speech setting.

3. Experimental setup

Rather than using a full prose section, Table 1 summarizes the dataset split, base training configurations, and evaluation protocol used throughout the project.

Table 1: *Dataset, protocol, and base configurations used throughout the project.*

Item	Setting
Train / dev / test	20k / 2k / 2k utterances
Primary metric	Test-set WER
Model families	Whisper, Wav2Vec2
Whisper models	tiny.en (39M), base.en (74M)
Wav2Vec2 model	wav2vec2-base-960h (95M)
Age bins	4–6, 7–9, 10–14, all
Task 3 Whisper base point	LR 5×10^{-6} , BS 16, 500 steps
Task 3 W2V base point	LR 1×10^{-5} , BS 8, 1000 steps
Best later-task Whisper cfg	LR 5×10^{-6} , BS 16, 500 steps
Best later-task W2V cfg	LR 3×10^{-5} , BS 8, 2000 steps
W2V fine-tuning note	stabilized setup with frozen feature encoder
Model selection	tuned on dev/validation, reported on test
Matched-exposure reruns	recomputed <code>max_steps</code> by subset size

Note: The age-group subsets were not perfectly balanced: the 10–14 subset contained more training examples than the 4–6 and 7–9 subsets. This mattered in Tasks 4 and 5 because fixed `max_steps` gave smaller subsets more effective epochs, which could make them appear stronger than they really were under a fair comparison.

4. Experiments and Results

4.1. Task 1: Whisper-base vs. Whisper-tiny

The goal of Task 1 was to compare how model size affects zero-shot transfer and fine-tuning gains for Whisper on children’s speech.

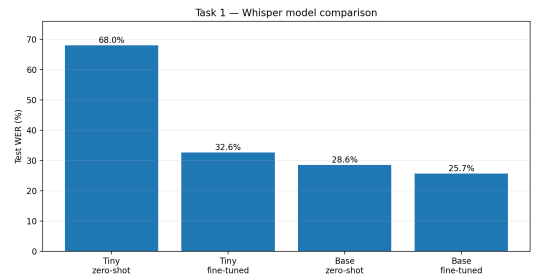


Figure 1: *Task 1 results: Whisper-tiny vs. Whisper-base in zero-shot and fine-tuned settings.*

Table 2: Salient outcomes from Task 1.

Model	Zero-shot WER	Fine-tuned WER	Δ WER
Whisper-tiny	0.6805	0.3263	-0.3542
Whisper-base	0.2856	0.2572	-0.0284

Whisper-tiny benefited much more from fine-tuning than Whisper-base because it began from a much weaker zero-shot starting point. In zero-shot mode, Whisper-base was already reasonably well aligned to the OGI Kids domain, so there was less room for improvement; fine-tuning only needed to make a relatively small correction from 0.2856 to 0.2572 WER. Whisper-tiny, by contrast, started at 0.6805 WER, which suggests a much larger mismatch between its pretrained representations and the acoustics of children’s speech. Fine-tuning therefore had a much bigger job to do: it was not only adapting the model to the dataset, but also correcting a weaker initial representation of pronunciation, speaking rate, and variability in child speech. This is why the absolute gain for Whisper-tiny was far larger. However, even after fine-tuning, Whisper-tiny still did not match Whisper-base, which suggests that the larger model’s extra capacity and stronger pretraining remained important even after adaptation.

4.2. Task 2: Wav2Vec2 zero-shot vs. fine-tuned

The goal of Task 2 was to measure how much Wav2Vec2 benefits from task-specific adaptation to children’s speech.

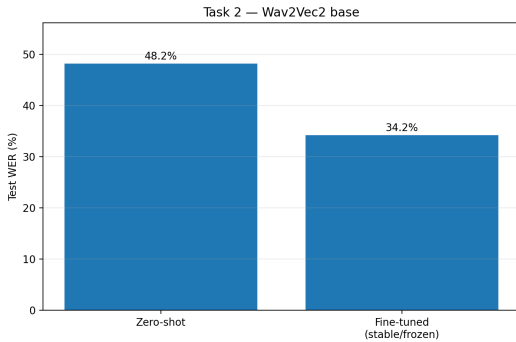


Figure 2: Task 2 results: Wav2Vec2-base in zero-shot and fine-tuned settings.

Table 3: Salient outcomes from Task 2.

Model	Zero-shot WER	Fine-tuned WER	Δ WER
Wav2Vec2-base	0.4821	0.3423	-0.1398

Fine-tuning helped Wav2Vec2 much more than it helped Whisper-base because Wav2Vec2 started from a weaker zero-shot baseline and therefore had a larger domain mismatch to correct. In zero-shot mode, Wav2Vec2 was less well aligned to the acoustics and pronunciation patterns of children’s speech, so supervised adaptation on OGI Kids produced a much larger gain. By contrast, Whisper-base already transferred relatively well before any task-specific training, so its fine-tuning improvement was naturally smaller. It is also worth noting that this was not a completely unrestricted fine-tune: in our stabilized setup, the Wav2Vec2 feature encoder remained frozen, so only part of the model was updated. Even with that limitation,

the performance gain was still substantial, which suggests that Wav2Vec2 had significant room to benefit from adaptation to the child-speech domain.

4.3. Task 3: Hyperparameter sweep

The goal of Task 3 was to run a parameterized sweep and choose a good training configuration for later tasks.

Figure 3 illustrates the search strategy. We varied one hyperparameter at a time while keeping the other two fixed, so the effect of learning rate, batch size, and training duration could be interpreted cleanly.

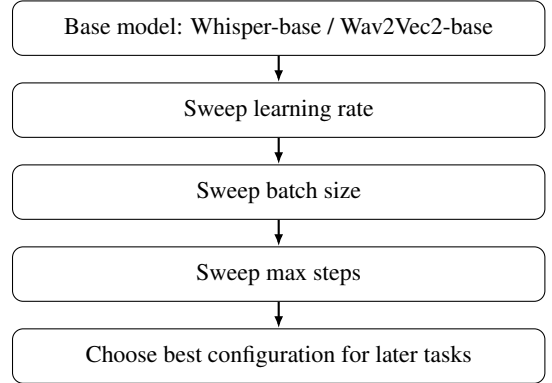


Figure 3: Task 3 search strategy: sweep one hyperparameter at a time, then choose the best configuration for later tasks.

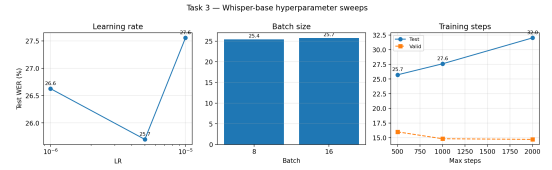


Figure 4: Task 3 Whisper hyperparameter sweep.

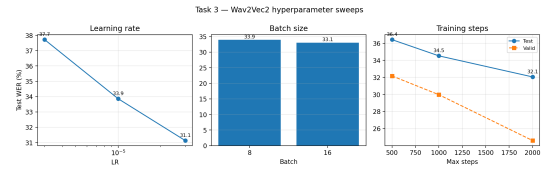


Figure 5: Task 3 Wav2Vec2 hyperparameter sweep.

The Task 3 sweeps show that the two model families responded differently to fine-tuning. For Whisper, the results suggest that a relatively careful fine-tuning setup worked best overall, and increasing the number of training steps too far could hurt test WER, suggesting overfitting or drift away from a strong pretrained starting point.

For Wav2Vec2, the trend was different: the model benefited more from a higher learning rate, which suggests it needed a stronger update signal to adapt to children’s speech. Batch size 8 performed best for both models. Overall, Task 3 identified the optimization trends that guided the configurations carried forward into later tasks.

Table 4 lists the final parameter choices carried forward to later tasks.

Table 4: *Final Task 3 configurations used for later experiments.*

Model	LR	Batch size	Max steps
Whisper-base	5×10^{-6}	16	500
Wav2Vec2-base	3×10^{-5}	8	2000

4.4. Task 4: Training-set size

The goal of Task 4 was to test the effect of training-set size on WER.

In the original version of Task 4, `max_steps` was held fixed across 5k, 10k, and 20k subsets. This created a confound: smaller subsets were iterated over more times, so they received more effective epochs per example. Under that setup, smaller subsets could appear surprisingly strong even when the comparison was not fully fair.

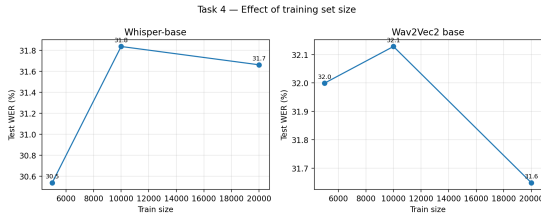


Figure 6: *Original Task 4 size sweep with fixed max_steps.*

After interpreting those results, we debugged the issue and traced it to the fixed-step setup. We then reran Task 4 with matched exposure for Wav2Vec2, recomputing `max_steps` so that each subset saw approximately the same number of effective passes through the data as the full-data reference configuration. Whisper did not produce usable rerun test results because its matched-exposure evaluation path appeared to break.

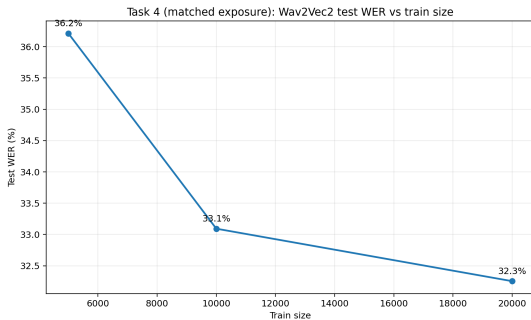


Figure 7: *Updated Task 4 rerun for Wav2Vec2 with matched exposure.*

In the corrected rerun, the trend became much more logical: for Wav2Vec2, more training data clearly produced lower WER. This strongly suggests that the original flat or counterintuitive Task 4 pattern was driven mainly by the fixed-step confound. Because the matched-exposure Whisper rerun did not produce usable test results, we do not draw a firm corrected Task 4 conclusion for Whisper.

4.5. Task 5: Age-group transfer

The goal of Task 5 was to study how training and testing on different age groups affects performance.

Again, the original experiment held `max_steps` fixed. That skewed the results by giving smaller age-specific subsets, such as ages 4–6 or 7–9, more effective epochs than larger subsets like 10–14 or all ages. In other words, the age-group comparison was partly mixing two effects: age-related difficulty and unequal training exposure.

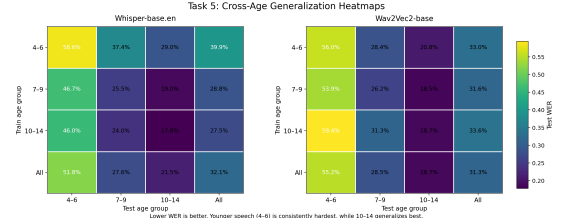


Figure 8: *Original Task 5 heatmaps with fixed max_steps.*

This imbalance made the original fixed-`max_steps` setup especially misleading, since smaller age-group subsets were repeated more times during training. We corrected this by scaling `max_steps` according to the size of each training subset.

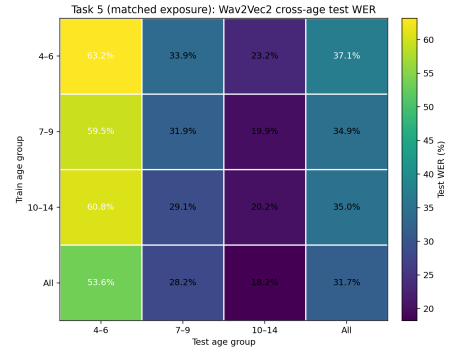


Figure 9: *Updated Task 5 rerun for Wav2Vec2 with matched exposure.*

The matched-exposure rerun again broke for Whisper, but it worked for Wav2Vec2 and clarified the result. The heatmap structure is mainly column-driven: the test age group dominates the difficulty, with ages 10–14 easiest and ages 4–6 hardest. After the correction, training on all ages produced the best overall cross-age performance. This replaced the earlier impression that narrow age-band training could sometimes be better.

4.6. Task 6: Data augmentation

Task 6 tested whether augmentation improved robustness. The original pipeline used simple waveform-level augmentation consisting of additive Gaussian noise and random gain scaling.

That setup hurt both models: Wav2Vec2 degraded from 0.3091 to 0.3248 WER, and Whisper degraded from 0.3213 to 0.3687.

We then reran Task 6 with more literature-aligned augmentation, specifically speed perturbation and vocal tract length perturbation (VTLP), which are standard augmentation strategies in prior ASR work [6, 7].

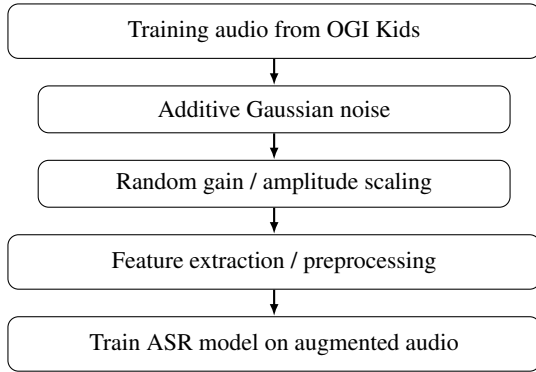


Figure 10: Original augmentation pipeline used in the first Task 6 experiment. Augmentation was applied only to the training audio and consisted of additive Gaussian noise followed by random gain scaling before feature extraction and model training.

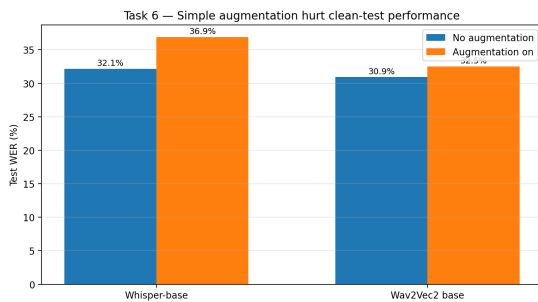


Figure 11: Original augmentation experiment: Gaussian noise + gain scaling.

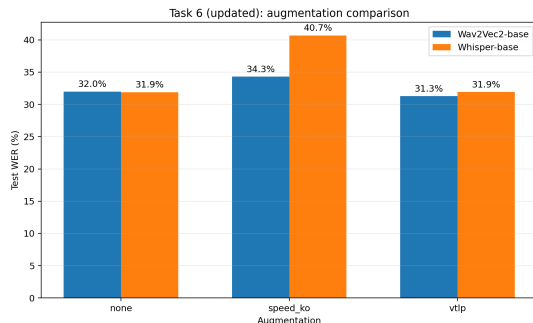


Figure 12: Updated augmentation rerun: none vs. speed perturbation vs. VTLP.

The overall conclusion for Task 6 is therefore not that augmentation solved the problem, but that augmentation was highly method-dependent and provided only limited benefit in this setup.

5. Summary and Discussion

Overall, the corrected rerun results strengthen two main conclusions. First, the original fixed-max_steps setup had partly distorted the Task 4 and Task 5 comparisons by giving smaller subsets more effective epochs. Second, after correcting for that confound, the most reliable Wav2Vec2 results showed that more data helps and that training on all ages gives the best overall cross-age generalization, while the dominant source of difficulty remains the youngest test group. Table 5 summarizes the main findings across the project.

Table 5: Key findings across tasks.

Task	Main empirical result	Interpretation
1	Whisper-base outperformed Whisper-tiny in zero-shot and after fine-tuning.	Larger model scale and stronger pretraining mattered; Whisper-tiny needed much more adaptation.
2	Wav2Vec2 improved strongly after stabilized fine-tuning.	Wav2Vec2 started from a larger domain mismatch and benefited more from supervision.
3	Hyperparameters strongly affected WER.	Whisper preferred careful fine-tuning; Wav2Vec2 preferred a higher LR and smaller batch size.
4	Original size sweep was confounded; matched-exposure W2V rerun showed more data helps.	Fixed max_steps had favored smaller subsets by giving them more effective epochs.
5	Age-group transfer was dominated by test age; matched-exposure W2V rerun showed all-age training worked best.	Younger speech was intrinsically harder, and fair exposure changed the training-age-group comparison.
6	Simple augmentation hurt; VTLP helped only slightly for W2V.	Augmentation must be matched carefully to the child-speech domain.

6. Future work

If we had more time, we would focus on three next steps that would most improve confidence in the current findings. First, we would run repeated-seed experiments and broader hyperparameter sweeps to improve confidence in close comparisons and test whether better configurations exist. This would also enable a more rigorous statistical significance analysis using confidence intervals or paired tests on per-utterance outputs. Second, we would explore more targeted augmentation strategies, especially child-speech-relevant ones. Third, we would perform a more detailed error analysis by breaking WER into substitutions, deletions, and insertions across age groups and inspecting representative transcription failures. It would also be interesting to test whether repeated exposure to smaller subsets can improve WER more than conventional augmentation, since this was closely tied to the fixed-max_steps confound in Tasks 4 and 5.

7. References

- [1] K. Shobaki, J.-P. Hosom, and R. Cole, “The OGI kids’ speech corpus and recognizers,” in *Proceedings of the International Conference on Spoken Language Processing*, 2000.
- [2] Linguistic Data Consortium, “CSLU: Kids’ speech version 1.1,” LDC2007S18, 2007, linguistic Data Consortium.
- [3] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” *arXiv preprint arXiv:2212.04356*, 2022.
- [4] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Advances in Neural Information Processing Systems*, 2020.
- [5] D. S. Park, W. Chan, Y. Zhang, C.-C. Chiu, B. Zoph, E. D. Cubuk, and Q. V. Le, “SpecAugment: A simple data augmentation method for automatic speech recognition,” in *Proceedings of INTERSPEECH*, 2019.
- [6] N. Jaitly and G. E. Hinton, “Vocal tract length perturbation (vtlp) improves speech recognition,” in *Proc. ICML Workshop on Deep Learning for Audio, Speech and Language*, 2013.
- [7] T. Ko, V. Peddinti, D. Povey, M. L. Seltzer, and S. Khudanpur, “Audio augmentation for speech recognition,” in *Proc. Interspeech*, 2015, pp. 3586–3589.